

3 The kidney is an organ in the human body that is involved in blood filtration and the production of urine.

- (a) (i) The afferent arteriole leading to the glomerulus is wider than the efferent arteriole leaving the glomerulus.

Which describes the effect that this produces?

(1)

- ☐ **A** it increases pressure in the glomerulus for ultrafiltration
- ☐ **B** it increases water movement by osmosis into the Bowman's (renal) capsule
- ☐ **C** it reduces the risk of capillary damage, due to high pressure
- ☐ **D** it reduces water reabsorption from the proximal tubule

- (ii) In each hour, the kidneys will filter $6\,000\text{ cm}^3$ of blood.

How much blood will be filtered in 24 hours?

(1)

- ☐ **A** 124 dm^3
- ☐ **B** 144 cm^3
- ☐ **C** 144 dm^3
- ☐ **D** $1\,440\text{ cm}^3$

- (b) The table shows the relationship between the ADH concentration in blood, concentration of urine and the rate of urine production.

Percentage of maximum ADH concentration in blood (%)	Concentration of urine / a.u.	Rate of urine production / $\text{cm}^3 \text{ minute}^{-1}$
0	90	20.0
25	700	8.0
50	950	4.0
75	1050	1.5
100	1150	1.0

- (i) Calculate the volume of urine produced in **one** day with an ADH concentration of 75%.

Assume that the concentration of ADH stays constant throughout the day.

Give your answer in standard form.

(2)

Answer cm^3

(ii) Describe the effect of ADH concentration on the concentration of urine and the rate of urine production.

(2)

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(c) Explain how ADH affects the movement of water from the lumen of the collecting duct into the blood.

(3)

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(Total for Question 3 = 9 marks)